What is Sound?

Sound is a form of mechanical energy that travels through a medium (such as air, water, or solids) as vibrations of particles. These vibrations propagate as longitudinal waves, meaning the particles of the medium oscillate back and forth in the same direction as the wave travels.

When an object vibrates, it causes the surrounding medium's particles to move, creating areas of compression (particles pushed together) and rarefaction (particles spread apart). These alternating regions travel outward from the source as sound waves.

Key characteristics of sound waves:

- Medium required: Sound cannot travel in a vacuum because it needs particles to vibrate.
- Speed of sound: Varies depending on the medium (e.g., ~343 m/s in air at room temperature, ~1500 m/s in water).
- Wave type: Longitudinal mechanical wave.

Properties of Sound

Sound waves can be described and analyzed using specific physical properties. These determine how we perceive sound (pitch, loudness, quality) and how we measure it scientifically.

Frequency (f) : The number of complete cycles (vibrations) of the sound wave per second. Measured in Hertz (Hz).

$$f = \frac{Number\ of\ cycles}{Time\,(seconds)}$$

High frequency → high-pitched sound (e.g., a whistle).

- Low frequency → low-pitched sound (e.g., a drum).
- Human hearing range: 20 Hz – 20,000 Hz (20 kHz).

Example: If a guitar string vibrates 440 times in 1 second:

$$f = \frac{440}{1} = 440\ Hz$$

Wavelength (λ) : The distance between two consecutive compressions or rarefactions in the sound wave.

$$\lambda = \frac{Speed\ of\ sound\ \ (m/s)}{Frequency\,(Hz)}$$

Example: Speed of sound in air = 343 m/s, frequency = 440 Hz:

$$\lambda = \frac{343}{440} \approx 0.78\, m$$

Amplitude (A) : The maximum displacement of particles from their rest position in the medium. Measured in meters (m) for particle displacement, but in sound engineering, loudness is usually expressed in decibels (dB).

Perceptual meaning:

- Larger amplitude → louder sound.
- Smaller amplitude → softer sound.

Example: A soft voice may produce sound at ~40 dB, while a jet engine can exceed 120 dB.

Velocity (v) : The speed at which sound travels through the medium.

$$v = f \times \lambda$$

Typical speeds:

- Air (20°C): ~343 m/s
- Water: ~1500 m/s
- Steel: ~5960 m/s

Example: If frequency = 1000 Hz and wavelength = 0.343 m:

$$v = 1000 \times 0.343 = 343\, m/s$$

Phase (φ) : Describes the position of a point in the wave cycle, measured in degrees (°) or radians.

- In-phase: Two waves have peaks and troughs at the same positions → constructive interference (louder sound).
- Out-of-phase (180°): Peaks align with troughs → destructive interference (quieter or silent).

Example: Two speakers producing the same tone in phase will sound louder; if out of phase by 180°, they may cancel each other.

Types of Sound

A. Speech: Sound produced by humans for communication, using the vocal cords and resonating cavities in the mouth, nose, and throat.

- **Characteristics:**
    - Structured sequence of phonemes.

    - Highly variable in tone, pitch, and rhythm.
    - Contains both voiced sounds (vocal cords vibrate) and unvoiced sounds (no vibration).
- **Applications:**
    - Speech recognition systems.
    - Natural language processing (NLP).
    - Voice assistants (e.g., Alexa, Siri).

B. Music : Organized and harmonious combination of sounds, often following rhythmic and melodic patterns.

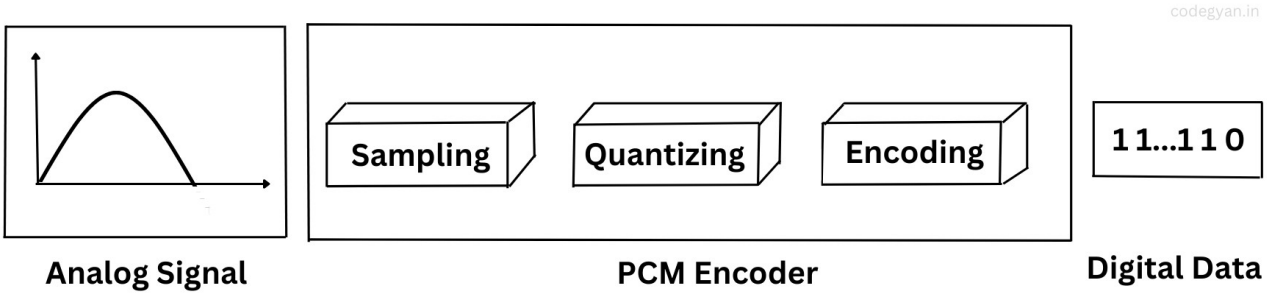
- **Characteristics:**
    - Pleasant to the ear (subjective but structured).
    - Based on notes, scales, and chords.
    - Contains harmonic relationships between frequencies.
- **Applications:**
    - Digital audio processing.
    - Music synthesis and MIDI systems.
    - Audio compression (MP3, AAC).

C. Noise : Unwanted, random, and unstructured sound that may be unpleasant or disturbing.

- **Characteristics:**
    - Random mix of frequencies.
    - No recognizable pitch or pattern.
    - Can mask other sounds, reducing clarity.
- **Applications:**
    - Noise reduction algorithms (e.g., in teleconferencing).
    - Signal filtering in audio engineering.
    - Acoustic environment analysis.

Unit 3

Sound in nature is analog — it is a continuous wave of air pressure variations that our ears detect. However, computers work with digital data — sequences of binary numbers (0s and 1s). Therefore, sound digitization is the process of converting a continuous sound wave into a discrete, numerical representation that can be stored, processed, or transmitted by a computer.



This process involves three main steps:

- Sampling – measuring the amplitude of the sound wave at regular intervals.
- Quantization – assigning each measured amplitude to the nearest value in a finite set of possible values.
- Encoding – representing the quantized values as binary numbers.

Sampling : It is the process of measuring the amplitude of the analog sound wave at fixed time intervals. The Sampling Rate (f_s) is the number of samples taken per second, measured in Hertz (Hz). According to the Nyquist–Shannon Theorem, to accurately represent the original sound:

$$f_s \geq 2 \times f_{max}$$

where f_{max} is the highest frequency present in the sound.

It mainly converts the continuous time axis into discrete time points.

Example: If human speech contains frequencies up to 8 kHz:

$$f_s \geq 2 \times 8{,}000 = 16{,}000\ Hz$$

So we choose fs=16 kHzfs=16kHz or higher for clear speech recording.

Quantization : It is the process of mapping each sampled amplitude to the nearest discrete level from a fixed set of possible values.

- The number of levels depends on the Bit Depth:

$$Levels = 2^{Bit\ depth}$$

- Higher bit depth → more precision, less quantization error → better sound quality.

It mainly converts the continuous amplitude axis into discrete values.

Example: If bit depth = 8 bits:

$$Levels = 2^8 = 256$$

This means each sample can be stored as one of 256 possible amplitude values.

Encoding : It converts each quantized value into its **binary representation** for storage or transmission.

- Each sample is stored as a binary number with a length equal to the bit depth.
- Example: A sample value of **150** (in decimal) becomes **10010110** in binary (8-bit encoding).

It mainly Transforms the discrete, quantized amplitude values into a computer-readable binary stream.

Advantages of Digitization

- High fidelity with proper sampling/bit depth.
- Easy to store, copy, and transmit without loss of quality.
- Enables advanced audio processing.

Representing Sound in Computers : Once sound is digitized (via sampling and quantization), it is represented as a sequence of binary numbers.

a) Sampling Rate

- The number of samples per second (in Hz).
- Higher sampling rate → captures more detail from the original sound.
- CD Audio: 44.1 kHz → 44,100 samples/sec.

b) Bit Depth (Sample Size)

- The number of bits used to store each sample.
- Higher bit depth → more amplitude precision, less quantization noise.
- CD Audio: 16 bits/sample.

c) Channels

- Mono: Single channel — same signal for all speakers.

- Stereo: Two channels — left and right audio signals stored separately.
- Surround: Multiple channels for 3D sound effects.

d) Encoding

- The binary representation of the quantized values.
- Can be uncompressed (raw PCM) or compressed (lossless/lossy formats).

How to choose the Right Audio Format?

Choosing the correct audio format depends on application requirements such as sound quality, storage space, and processing speed.

A. Uncompressed Formats

- Example: WAV (Waveform Audio File Format), AIFF (Audio Interchange File Format)
- Features: Store raw Pulse Code Modulation data without compression.
- Pros: Maximum quality, easy to edit.
- Cons: Large file sizes.
- Use Cases: Professional audio editing, studio recording.

B. Lossless Compressed Formats

- Example: FLAC (Free Lossless Audio Codec), ALAC (Apple Lossless Audio Codec)
- Features: Compress data without losing quality.
- Pros: Smaller size than uncompressed, identical sound quality.
- Cons: Larger than lossy formats.
- Use Cases: Archiving, audiophile listening.

C. Lossy Compressed Formats

- Example: MP3 (MPEG Audio Layer 3), AAC (Advanced Audio Coding)
- Features: Discard some audio data to reduce file size.
- Pros: Very small file sizes, good for streaming and portable devices.
- Cons: Lower sound quality (especially at low bit rates).
- Use Cases: Music streaming, podcasts, web audio.

Choosing Criteria: When deciding on a format, consider:

Quality Requirement:

- Professional editing → uncompressed or lossless.
- Casual listening → lossy at high bitrates.

Storage Space & Bandwidth:

- Limited storage → use compressed formats.

Playback Device Compatibility:

- Ensure the chosen format is supported by the target devices.

What is MIDI?

MIDI is a digital communication protocol introduced in 1983 to allow electronic musical instruments, computers, and other digital devices to communicate musical performance data.

- MIDI does not store or transmit actual sound.
- Instead, it transmits instructions describing:
  - Which note to play.
  - When to start and stop the note.
  - How loud or soft to play (velocity).
  - Additional parameters like pitch bend, modulation, instrument change.

Example: When you press a key on a MIDI keyboard, it doesn't send an audio waveform. Instead, it sends a message like:

$$Note\,On : Channel\,1\,, Note\,60\,(Middle\,C)\,, Velocity\,90$$

The receiving device (e.g., synthesizer or computer) generates the actual sound.

How does MIDI work?

MIDI transmits messages between devices. Each message has a specific structure:

A. Structure of a MIDI Message

- Status Byte: Identifies message type and channel (e.g., Note On, Note Off).
- Data Byte 1: Note number (0–127; Middle C = 60).
- Data Byte 2: Velocity (0–127; how hard the note is played).

Example: $0\,x\,90\,(Note\,On\,, channel\,1) \vee 60\,(Middle\,C) \vee 90\,(Velocity)$

B. Communication Method

- MIDI is serial communication at 31.25 kbps.
- Uses 16 channels — up to 16 instruments can be controlled in one MIDI connection.
- Can carry note events, control changes, program changes, and system messages.

C. MIDI Data Flow

- Input Device: Keyboard, drum pad, MIDI guitar.
- MIDI Cable/Interface: Transmits the data.
- Output Device: Synthesizer, sound module, or computer.
- Sound Generation: Output device plays sound according to received instructions.

Digital Audio vs MIDI

Digital Audio	MIDI
Digital Audio refers to the reproduction and transmission of sound stored in a digital format.	A MIDI is a software for representing musical information in a digital format.
Digital Representation of physical sound waves.	Abstract Representation of musical sound and sound effects.
Digital Audio comprises analog sound waves that are converted into a series of 0s and 1s.	MIDI comprises a series of commands that represent musical notes, volume, and other musical parameters.
Actual Sound is stored in a digital audio file.	No actual sound is stored in the MIDI file.
Files are large in size and are loose.	Files are small in size and compact.
The quality of sound is in proportion to the file size.	The quality of sound is not in proportion to the file size.
They reproduce the exact sound in a digital format.	They sound a little different from the original sound.
Digital audio is used for recording and playback of music, sound effects, and voice-overs.	MIDI is used for creating and controlling electronic music, such as synthesizers and drum machines.